32K contexts · shared SAE features by task and layer

Current Q and high-attention historical K share at least one feature

New: 100 contexts · Top 1% · No Q–K matching

	3	7	11	15	19	23	27	31	35	39	43	47	51	55	59	63
UUID retrieval	29	29	30	37	41	34	30	33	35	32	31	31	35	24	25	29
Nested JSON	29	29	31	38	40	36	30	37	37	34	31	30	33	26	28	30
HTML to TSV	28	28	29	39	45	35	31	35	43	39	36	34	38	28	28	29
Code patch	28	28	29	35	36	33	29	35	38	40	35	33	35	29	29	30
Staged evidence	28	28	30	38	41	36	29	35	36	34	32	33	35	26	27	31
ToM tracking	30	30	35	47	47	54	47	54	56	58	50	43	45	37	34	34
Countdown	34	32	37	45	46	54	51	57	62	58	58	52	47	39	35	38

Old: 7 contexts · Top 1% · No Q–K matching

	3	7	11	15	19	23	27	31	35	39	43	47	51	55	59	63
UUID retrieval	29	29	30	37	40	34	29	32	34	30	31	29	34	24	25	28
Nested JSON	29	29	31	37	39	35	30	37	37	32	31	29	33	26	28	30
HTML to TSV	28	28	30	40	45	36	33	35	41	38	36	33	38	28	27	29
Code patch	28	28	29	35	36	33	29	35	37	38	33	29	34	29	28	31
Staged evidence	28	28	30	38	41	35	30	36	36	34	33	34	36	27	27	30
ToM tracking	30	30	33	44	45	54	47	56	56	57	47	42	46	38	35	33
Countdown	35	32	36	44	45	54	52	57	62	60	60	53	46	37	34	38

New: 100 contexts · Top 1% · With Q–K matching

	3	7	11	15	19	23	27	31	35	39	43	47	51	55	59	63
UUID retrieval	42	37	35	39	56	38	55	68	75	63	66	64	63	77	76	62
Nested JSON	43	39	38	43	54	45	59	67	72	61	67	64	64	75	76	62
HTML to TSV	43	38	35	43	58	43	55	71	77	68	68	63	68	73	72	60
Code patch	43	39	38	45	52	48	59	67	72	69	71	64	64	77	76	63
Staged evidence	43	38	36	41	56	41	56	67	74	63	68	65	65	77	77	62
ToM tracking	53	46	51	71	71	76	76	84	83	83	76	70	70	79	78	66
Countdown	61	48	54	62	65	73	81	89	90	88	87	80	75	86	85	75

Old: 7 contexts · Top 1% · With Q–K matching

	3	7	11	15	19	23	27	31	35	39	43	47	51	55	59	63
UUID retrieval	42	38	35	40	55	38	55	67	72	60	65	61	61	76	74	61
Nested JSON	43	39	37	43	53	45	59	68	73	61	67	62	63	75	74	61
HTML to TSV	43	39	35	44	57	45	56	72	77	67	69	64	69	75	73	61
Code patch	42	38	37	45	51	46	59	69	73	69	71	64	64	75	74	62
Staged evidence	42	37	35	41	56	41	56	65	74	61	67	64	63	76	76	62
ToM tracking	53	47	52	69	68	76	74	80	77	78	71	63	63	73	75	60
Countdown	61	49	54	62	64	73	81	89	90	90	88	80	75	87	85	75

New: 100 contexts · Top 2% · No Q–K matching

	3	7	11	15	19	23	27	31	35	39	43	47	51	55	59	63
UUID retrieval	24	24	25	31	34	28	25	27	29	27	27	27	31	24	24	25
Nested JSON	24	24	25	31	32	29	25	29	30	28	27	26	29	25	26	26
HTML to TSV	24	24	24	33	38	29	27	30	36	33	31	30	33	27	27	25
Code patch	23	23	23	28	29	27	24	29	32	34	31	29	30	27	27	26
Staged evidence	23	23	25	32	34	29	24	29	31	28	28	29	31	25	26	26
ToM tracking	24	25	28	38	38	46	40	47	50	52	44	37	39	35	32	29
Countdown	27	26	29	36	37	43	42	48	54	51	51	44	40	36	34	32

Old: 7 contexts · Top 2% · No Q–K matching

	3	7	11	15	19	23	27	31	35	39	43	47	51	55	59	63
UUID retrieval	24	24	25	31	33	28	25	26	28	25	27	25	30	23	24	25
Nested JSON	24	24	25	30	31	28	25	30	30	27	27	26	29	25	26	25
HTML to TSV	23	24	25	33	38	30	28	30	35	32	31	29	33	27	26	25
Code patch	23	23	23	28	29	26	24	29	31	32	29	26	30	27	26	26
Staged evidence	23	23	25	32	34	28	26	30	31	29	29	29	32	26	26	26
ToM tracking	24	24	26	35	37	45	41	49	51	52	42	37	40	36	33	29
Countdown	28	26	29	35	36	43	43	48	54	52	53	45	39	35	33	32

New: 100 contexts · Top 2% · With Q–K matching

	3	7	11	15	19	23	27	31	35	39	43	47	51	55	59	63
UUID retrieval	32	29	28	31	43	29	41	52	61	49	54	50	51	60	58	48
Nested JSON	33	30	30	33	41	33	44	51	58	46	53	49	50	58	58	49
HTML to TSV	33	29	28	34	45	33	43	57	63	54	57	51	56	57	56	48
Code patch	33	30	29	34	39	35	45	53	59	56	58	51	52	61	59	49
Staged evidence	33	29	28	32	43	31	42	52	60	49	55	51	52	60	59	49
ToM tracking	40	36	41	59	58	64	65	76	75	75	69	61	60	67	65	54
Countdown	49	39	43	50	51	58	69	81	83	81	79	69	64	75	72	64

Old: 7 contexts · Top 2% · With Q–K matching

	3	7	11	15	19	23	27	31	35	39	43	47	51	55	59	63
UUID retrieval	33	30	28	31	42	29	41	51	59	46	52	48	49	58	57	48
Nested JSON	33	30	29	32	40	33	44	51	58	46	53	48	49	57	57	48
HTML to TSV	33	30	28	35	45	34	44	58	63	54	57	52	57	59	57	49
Code patch	32	29	28	34	38	35	45	54	59	56	58	50	51	59	57	48
Staged evidence	32	29	28	32	44	31	43	50	60	48	54	50	51	59	59	49
ToM tracking	40	36	41	57	56	64	64	73	69	71	63	54	54	62	62	50
Countdown	49	39	43	49	50	58	69	81	83	83	80	69	64	75	72	64

Standard attention layer (zero-based)

Coverage (%)

The top fraction is ranked by exact Full attention over the prompt. New cells contain 14–15 contexts per task. Old cells contain one.